Rana Radiansyah

Assignment 2

PSYCH 403

EEG Racing Caterpillar: Concept, Implementation, Challenges, and Learning.

The EEG Racing Caterpillar project is a visual, interactive interpretation of brainwave activity, designed to make abstract neural signals tangible and engaging. Each caterpillar represents a standard EEG frequency band, Alpha, Beta, Gamma, Theta, and Delta, and moves along a track according to its corresponding brainwave intensity. By mapping EEG signals to visual parameters such as speed, color, and motion, the project transforms complex, abstract data into a dynamic, intuitive experience.

Brain activity can be divided into frequency bands that correspond to distinct cognitive or physiological states. For example, Alpha waves are more prominent during relaxed and calm states, Beta waves increase during active thinking and concentration, Gamma waves are associated with high-level cognitive processing, Theta waves correspond to deep meditation, drowsiness, and creativity, and Delta waves dominate during deep sleep. By translating these signals into animated caterpillar movements, the project provides a playful yet informative visualization of which brain activity is more dominant at a given moment.

The project is grounded in the principles of biofeedback and signal mapping. While EEG data can be collected from devices such as Muse, this project uses simulated EEG data from BrainImation. Signals are filtered into standard frequency bands, and each caterpillar's movement, color, size, and wiggle amplitude are updated in real time to reflect the EEG intensity. Caterpillars race towards a green finish line, creating a clear visual representation of the relative activity of each brainwave.

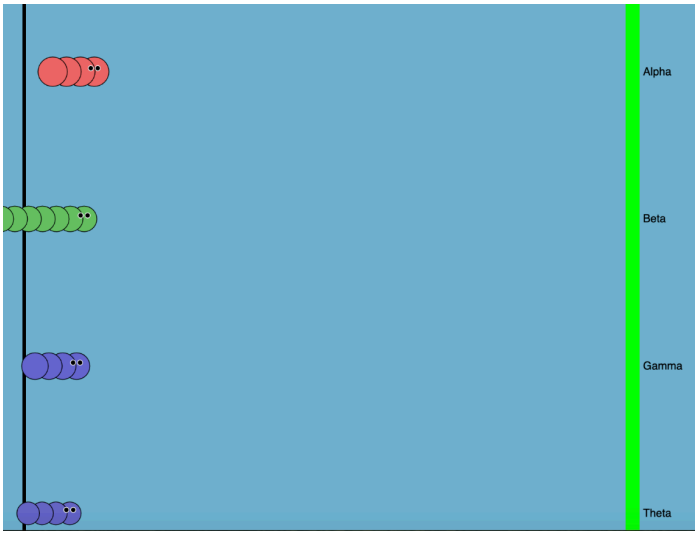
During development, I faced several challenges. As someone with limited coding experience, I had to navigate the differences between p5.js and BrainImation. For example, I initially used circle() to draw caterpillar segments, which did not work in BrainImation; switching to ellipse() resolved this. Additionally, while createCanvas() is required in p5.js, BrainImation handles the canvas automatically. I also attempted to implement a mouse-click feature to start the race dynamically, but integrating it correctly with BrainImation proved difficult. These challenges highlighted the importance of attention to detail, debugging, and learning platform-specific requirements. Despite these obstacles, the process expanded my understanding of programming and interactive visualization.

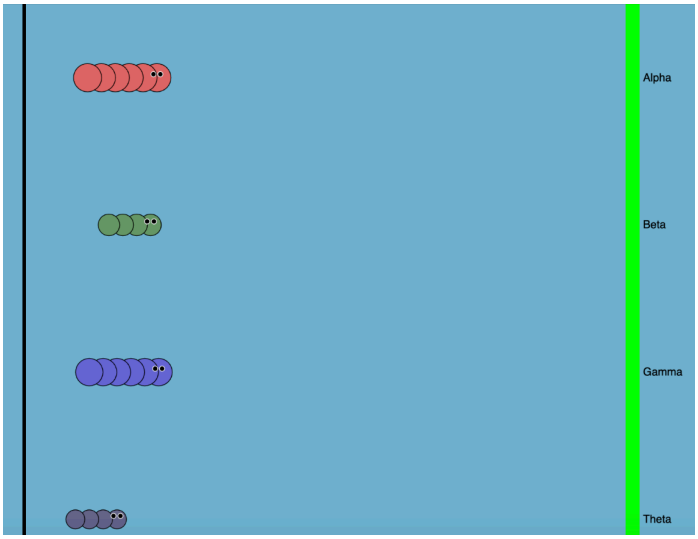
This project also provided significant learning opportunities. I gained insight into how brain activity can be quantified and interpreted in real time and how multiple visual parameters, speed, size, color, and motion, can be combined to represent data intuitively. I learned the importance of scaling, smooth transitions, and proportional representation to make visualizations both understandable and aesthetically engaging. I also discovered that small changes in code can have large perceptual effects, reinforcing the value of careful design and iterative testing.

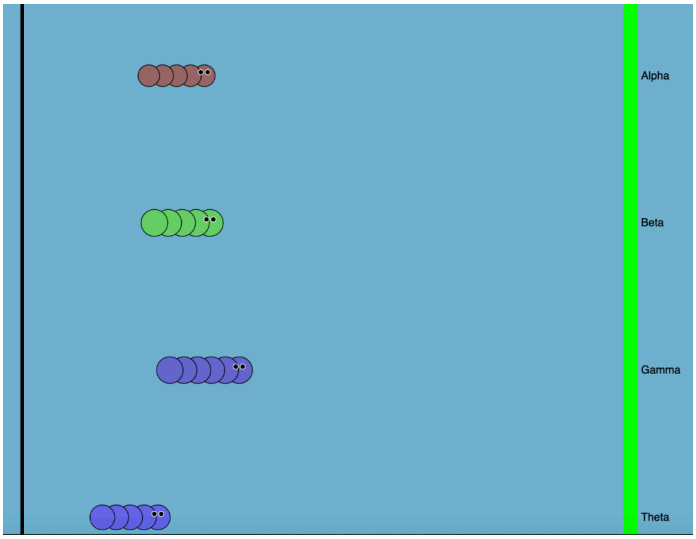
Finally, the project helped me practice programming in a modular, systematic way. By building functions for drawing caterpillars, moving them, and detecting winners, I developed a structured approach to coding and visual problem solving. Encountering errors and debugging strengthened my attention to detail and analytical skills, while observing the caterpillars' dynamic behavior made the learning process enjoyable and interactive.

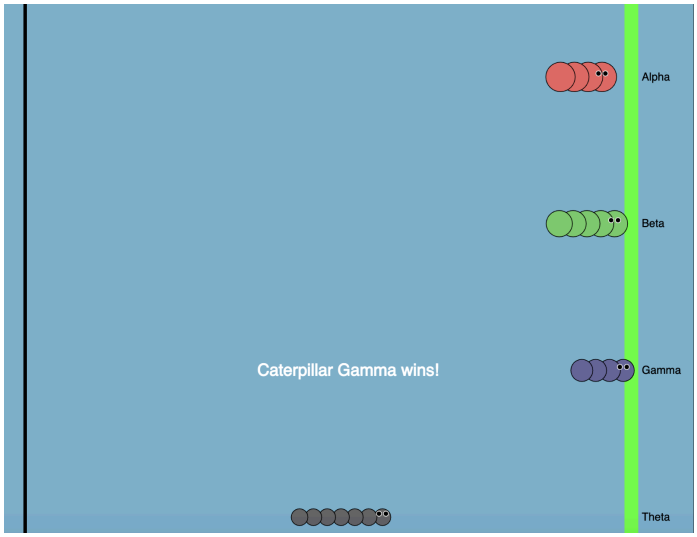
In conclusion, the EEG Racing Caterpillar project has been a valuable experience that bridges neuroscience, programming, and visual design. It demonstrates how abstract brainwave data can be transformed into an engaging, intuitive visualization, fostering both understanding and creativity. While there is still room for refinement and expansion, this project has enhanced my technical skills, deepened my understanding of cognitive science principles, and inspired me to continue exploring the intersection of neuroscience, technology, and interactive design.

Here are some images from the simulation:






Alpha
Beta
Gamma
Theta


Alpha
Beta
Caterpillar Gamma wins!
Gamma
Theta